

6-The impulse response $h(n)$ of a linear time invariant system is given by:

$$h(n) = u(n+3) + u(n-2) + 2u(n-7), \text{ where } u(n) \text{ is the unit step sequence.}$$

The above system is:?

- (a) stable and causal b. stable but not causal c. causal but unstable
d. unstable and not causal e. None

7-Determine the product of two signals: $x_1(n) = \{2, 1, 1.5, 3\}$; $x_2(n) = \{1, 1.5, 0, 2\}$.

- (a) $\{2, 1.5, 0, 6\}$ b. $\{2, 1.5, 6, 0\}$ c. $\{2, 0, 1.5, 6\}$ d. $\{2, 1.5, 0, 3\}$

8- $h_1(t)$ convolved with $h_2(t) = h_2(t)$ convolved with $h_1(t)$

- (a) Associative property b. Distributive property
c. Commutative property d. Multiplication property

9- find $y(n) = x(n) * h(n)$, where $x(n) = [1, 1]$ and $h(n) = [1, 1]$

- a. $[1, 2, 1]$ b. $[1, 2, 3]$ c. $[3, 1, 1]$ d. $[1, 1, 2]$

10- Find the fundamental period and frequency of the following signal

$$x(t) = 2\cos(6\pi t + \pi/6)$$

- a. $T=0.16$ sec, $f=6$ Hz b. $T=10$ sec, $f=0.1$ Hz c. $T=0.33$ sec, $f=3$ Hz d. $T=5$ sec, $f=0.2$ Hz

11- Is the following signal an energy signal

$$x(t) = u(t) - u(t-1) + t(t-4)$$

(a) YES

b. NO

12- Find the value of $2h[n] * \delta(n+1)$

- a. $2h[n-2]$ b. $2h[n]$ c. $2h[n-1]$ d. $2h[n+1]$

13- $h_1(t) * h_2(t) * h_3(t) = h_3(t) * h_2(t) * h_1(t)$

this property of convolution process is called _____

- a. Associative property
☒ b. Commutative property
c. Distributive property
☒ d. convolved with impulse property.

14- Noise generated by an amplifier of radio is an example for?

- a. Discrete signal
b. Deterministic signal
☒ c. Random signal
d. Periodic signal

15- any time delay or time in advance in the input signal leads to a different time delay or time in advance in the output signal :

in this case the system is called

- ☒ a. Time invariant systems
b. Memory less systems
c. Causal systems
d. message frequency

Question	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
Answer	C	e	C	d	b	A	o	A		C
Question	11	12	13	14	15					
Answer	o	d	b	C	A					

1-given $x(t)=-3t+15$ where, $4 \leq t \leq 5$. find the time interval for $x(-2t+4)$, Select one:

- a. $-0.8 \leq t \leq -0.4$ b. $-1.75 \leq t \leq -1.5$ c. $-0.75 \leq t \leq -0.5$ d. $0.5 \leq t \leq 0.75$ e. None
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2-given $x(t)=t+4$ where $-4 \leq t \leq -2$, find $x(-2t+4)$

- a. $x(t)=-4t+4$ where, $1 \leq t \leq 1.5$ b. $x(t)=-4t+6$ where, $1 \leq t \leq 1.5$
c. $x(t)=-4t+4$ where, $1.5 \leq t \leq 2.5$ d. $x(t)=-4t+6$ where, $1.5 \leq t \leq 2.5$ e. None
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3-if $x(t)=r(t+3)-r(2t)$, the $x(t)$ during the time intervals $-3 \leq t \leq 0$ can be written as

- a. 3 b. $t+3$ c. $3-t$ d. $t-3$ e. None
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4- $x(t)=r(t)-2r(t-2)$ find $x(-2t+2)$.

- a. $x(-2t+2)=2-2t$ ($-1 < t < 0$), $x(-2t+2)=2t+2$ ($0 < t < 1$)
b. $x(-2t+2)=2-2t$ ($0 < t < 1$), $x(-2t+2)=2t+2$ ($-1 < t < 0$)
c. $x(-2t+2)=\underline{2t+2}$ ($0 < t < 1$), $x(-2t+2)=2t+2$ ($-2 < t < 0$)
d. $x(-2t+2)=-2t+2$ ($0 < t < 1$), $x(-2t+2)=2t+2$ ($-1 < t < 0$)
e. None
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5-If $x(n)=\overset{0}{(0,0,1,2,3,4,0,0)}$ then $x(n-3)$ is

- a. $(0,0,0,2,4,6,8,0,0)$ b. $(0,0,0,1,2,3,4,0,0)$ c. $(1,2,3,4,0,0,0,0,0)$ d. $(0,0,0,0,0,1,2,3,4)$
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